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Prop and Display Naming Conventions

These conventions ensure consistent naming across:

- **LOR sequencing (Channel Grid)**
- **Display database records (Postgres / SQLite)**
- **Physical labels used in the field**
- **Field wiring exports**

If naming rules are not followed, channels scatter in the grid, wiring reports drift from reality, and physical labels no longer match database records.

This document prevents chaos.

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1. Two Separate Naming Layers in LOR

LOR uses two different fields that serve different purposes:

Field	Purpose	Spaces Allowed	Used as Database Key
Name	Channel grid sorting & programming	Yes	No
Comment	Physical display identifier	No	Yes

These fields must not be confused.

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2. LOR Name Field (Channel Naming Convention)

The **Name field controls alphabetical sorting inside the LOR grid.**

LOR sorts strictly alphabetically.
Improper numbering or inconsistent formatting causes channels to scatter.

2.1 Required Structure

Recommended format:

```
<LL> <Group> <UID>-<CH> <Description>
```

Example:

```
TC 7B-01 Hippo Box
```

2.2 UID and Channel Rules (CRITICAL)

UID

- Controller ID in HEX
- Uppercase
- No leading zero padding
- Example: `7B`, `1F`, `20`

Channel

- Always 2-digit padded
- `01–16`
- Even if wiring exports show a single digit

Correct:

- `7B-01`
- `7B-02`
- `7B-10`

Incorrect:

- `7B-1`
- `7B-2`

If channels are not padded, alphabetical sorting breaks:

Wrong order:

- 7B-1
- 7B-10
- 7B-11
- 7B-2

Correct order:

- 7B-01
- 7B-02
- 7B-03
- 7B-10

2.3 Real Example (Hippo + Carolers)

Channel Grid Example

This grid shows:

- One multi-state animated display (Hippo)
- Three physical panels (CarolerPanel-01..03)
- Proper grouping by padded channel numbers

The naming works because:

- All channels begin with consistent stage prefix
- Channels are padded
- Related elements sort together

2.4 Unused Channels (SPARE Rule — REQUIRED)

Any unused channel on a controller must be explicitly added to the preview.

Do NOT leave gaps.

Unused channels must:

- Follow the exact same naming structure
- Use the same Stage and Group prefix
- Use correct HEX UID
- Use 2-digit padded channel number
- End with the word `Spare`

Example:

TC 7B-11 Spare
TC 7B-12 Spare

Why this is required:

- Makes unused capacity immediately visible in the Channel Grid
- Prevents accidental reuse of a channel already assumed to be empty
- Keeps controllers fully documented
- Allows quick identification of expansion space during programming

Unused channels must remain in the preview even if no physical wiring exists.

When a Spare channel is later assigned to a display:

- Rename the channel
- Update wiring documentation
- Remove the `Spare` designation

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3. Comment Field (Display Identifier)

The **Comment field defines the Display Name** used for:

- Physical labels
- Database joins
- Wiring exports
- Inventory tracking
- Maintenance records

This is the true identity of the physical display.

3.1 Comment Field [Display Name] Rules

- No spaces allowed
- Must start with 2-letter stage abbreviation
- Use hyphen as structural separator
- Use a hyphen after the Actual Display Name (CamelCase)
- Must remain stable over time

Examples:

Standard stages:

- `FC-Arch-01`means Food Collection Arch #1
- `RA-Arch-DS-01`means Racing Arches, Driver Side #1
- `EC-Elf-P2-06`means Elf Choir, Elf, Pattern 2, #6

GG stage (Controller Unit ID's specifically allowed):

- `GG-Elden-20-01 means Glistening Grove, Elden (version 1), UID 20 #1
- `GG-Elden-20-04`means Glistening Grove, Elden (version 1), UID 20 #4
- `GG-EldenV2-30-01 means Glistening Grove, Elden (version 2), UID 30 #1
- `GG-EldenV2-30-02`means Glistening Grove, Elden (version 2), UID 30 #2

3.2 Displays vs Sub-Props

Term	Definition
Display	One physical panel or unit
Sub-Prop	Logical motion element within a display

All sub-props must share the **exact same Comment value** as their parent display.

Example:

Channel Name	Comment [Display Name]
`TC 7B-01 Hippo Box`	`TC-ChristmasHippo`
`TC 7B-04 Hippo Body Mid`	`TC-ChristmasHippo`
`TC 7B-05 Hippo Body Full Head`	`TC-ChristmasHippo`

3.3 Example – Multi-Panel Display (Caroler)

Visualization Example

Three physical displays (panels):

- `TC-CarolerPanel-01`
- `TC-CarolerPanel-02`

- `TC-CarolerPanel-03`

Each has its own Comment value. TC is the Stage Code for Traditional Christmas

Even if programming groups them visually, they remain separate physical units in inventory and wiring.

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4. Display Name Format (put into Comment field)

Display names must be structured so that:

The first hyphen separates Stage from DisplayName Subsequent hyphens separate functional attributes No attribute codes may be appended directly to DisplayName without a hyphen

Prohibited:

EntryArchWrapDS SteelEntryArchPS CarCounterArchGrn

Required (Stage Code-Display Name-Attributes):

- IT-EntryArchWrap-DS
- IT-SteelEntryArch-PS
- FC-CarCounterArch-Grn

`<Stage Code>-<DisplayName>-<Variation>-<Sequence>-<Color>`

Segment	Meaning
Stage	Two-letter stage code
DisplayName	CamelCase, no spaces
Variation	Optional location/version identifier
Sequence	Always 2-digit padded
Color	Optional suffix

More Examples:

- `EC-Elf-P2-06`
- `RA-Arch-DS-01`
- `WF-MiniTree-G-04`
- `GG-Elden-20-01`
- `DF-Wrap-DS-A-02G`

4.1 DeviceType = Undetermined (Inventory-Only Displays)

Undetermined DeviceType Example

In the LOR preview editor, DeviceType is set to:

Undetermined

During SQLite parsing and Postgres ingestion, this becomes:

DeviceType = None

Operationally, both represent the same thing:

A physical inventory item with no controller assignment and no channel grid usage.

`<Stage Code>-<DisplayName>-<Variation>-<Sequence>-<Color>`

Use Cases

- Physical duplicates of programmed displays
- Arch supports
- Wrap stands
- Frame stands
- Future expansion inventory
- Volunteer Trailer Steps
- Igloo wagon for No Left Turn
- Additional displays were we duplicate the controllers for programming simplification like Emojis

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Quantity Creation (Max Circuits per Unit)

When creating inventory-only items:

- Set "Max Circuits per Unit" equal to the required quantity.
- LOR automatically generates suffixed records:
  - `-01`, `-02`, `-03`, etc.

Example:

Comment: `FC-WrapStand`

Max Circuits per Unit: `32`

Generated records:

- `FC-WrapStand-01`
- `FC-WrapStand-02`
- ...
- `FC-WrapStand-32`

If Max Circuits per Unit = 1:

- No suffix is added.

⚠ **Critical Warning**

The default value for "Max Circuits per Unit" is 16.

If not changed intentionally, LOR will create 16 inventory records automatically.

This will:

- Inflate inventory counts
- Pollute the database
- Require manual cleanup

Always verify this value before saving.

Rules

Inventory-only entries:

- Must follow full Comment naming rules
- Must not contain Network, UID, or ChannelGrid
- Must represent real physical inventory

These entries appear in database ingestion but do not affect wiring exports.

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5. Field Wiring Alignment

Field Wiring Example

Field wiring exports rely on the Comment value.

If Comment naming is incorrect:

- Wiring sheets become inaccurate
- Displays cannot be located in the field
- Maintenance records break

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6. Inventory and Physical Labels

Every display label must:

- Match the Comment field exactly
- Be permanently attached to the display
- Match database and wiring documentation

The Comment value is the backbone of:

- Physical inventory
- Storage tracking
- Repair history
- Field setup instructions

☑ Summary

Field	Controls	Critical Rule
Name	Programming & Grid Sorting	Pad channels to 2 digits
Comment	Physical Identity	No spaces, stable, structured

Naming consistency prevents:

- Channel grid chaos
- Lost programming
- Wiring mismatches
- Inventory drift

> **Revision History**
>
> - GAL 26-03-24 — Added images back in and fixed some typos
> - GAL 26-03-17 — Clarified Section 4 due to label printing conflicts
> - GAL 26-02-22 — Major clarification: separated Name vs Comment logic, added UID padding rule, added real-world examples with grid and wiring screenshots
> - GAL 25-10-30 — Formatting cleanup
> - GAL 25-10-29 — Initial merge